Introduction to the mcst package.

Monte carlo portfolio risk with stress testing simulations.

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January 16, 2026

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1 Introduction

The `portfolio_risk_simulation` function is a self-contained, end-to-end analytical engine for multi-asset portfolio risk modelling. Given a historical return series for a set of assets, it estimates the joint return distribution, constructs a forward-looking covariance matrix, simulates thousands of portfolio return paths, and - if specificed - applies structured stress scenarios to quantify how shocks propagate across assets. Its design philosophy is that a risk practitioner should be able to move from raw returns data to a fully characterised risk distribution in a single function call.

1.1 Core Use Cases

- **Portfolio VaR and CVaR:** Simulate the full distribution of weekly or monthly portfolio returns and extract tail-risk metrics.
- **Distribution selection:** Choose between Gaussian, Student-t, or Skewed Generalised t (SGT) marginals, with or without a copula-based dependence structure.
- **Volatility regime awareness:**Choose between historical, EWMA, or GJR-GARCH conditional volatility for the forward-looking covariance.
- **Stress testing:** Apply mean and/or volatility shocks to individual portfolio assets (endogenous) or to external instruments (exogenous), and observe how the shock propagates through the portfolio correlation structure.
- **Diagnostic visualisation:** Inspect marginal distributions, copula fit, covariance structure, and tail dependence.

1.2 General Workflow

The function executes the following pipeline sequentially:

Table 1: Pipeline summary

Step	Label	Description
1	Data Preprocessing	Validate inputs, detect return frequency, compute summary statistics.
1.5	Calibrate Parameters	Fit distribution-specific parameters (df, skewness, tail shape) to each asset.
2	Weights Processing	Accept equal, custom, or sweep-weight matrices; set simulation draw count.
3	Fit Marginal Distributions	Fit Normal, t, or SGT marginals per asset; optionally transform to uniform PIT scores.
4	Forward-Looking Volatility	Estimate forward variances via sample covariance, EWMA, or GJR-GARCH.
5	Fit Copula	Fit Gaussian or t-copula to the PIT residuals; extract correlation and tail dependence.
6	Build Covariance Matrix	Combine forward vol (D) and copula correlation (rho) into Sigma = D rho D.
6.5	Latent Diagnostics	Compare latent (uniform) space to observed space; run KS uniformity test.
7	Stress Testing	Apply endogenous and/or exogenous shocks; propagate via Schur complement.
8	Simulate Returns	Draw from the fitted joint distribution; compute portfolio return distributions.
9/10	Statistics & Plots	Compute risk metrics; produce diagnostic visualisations.

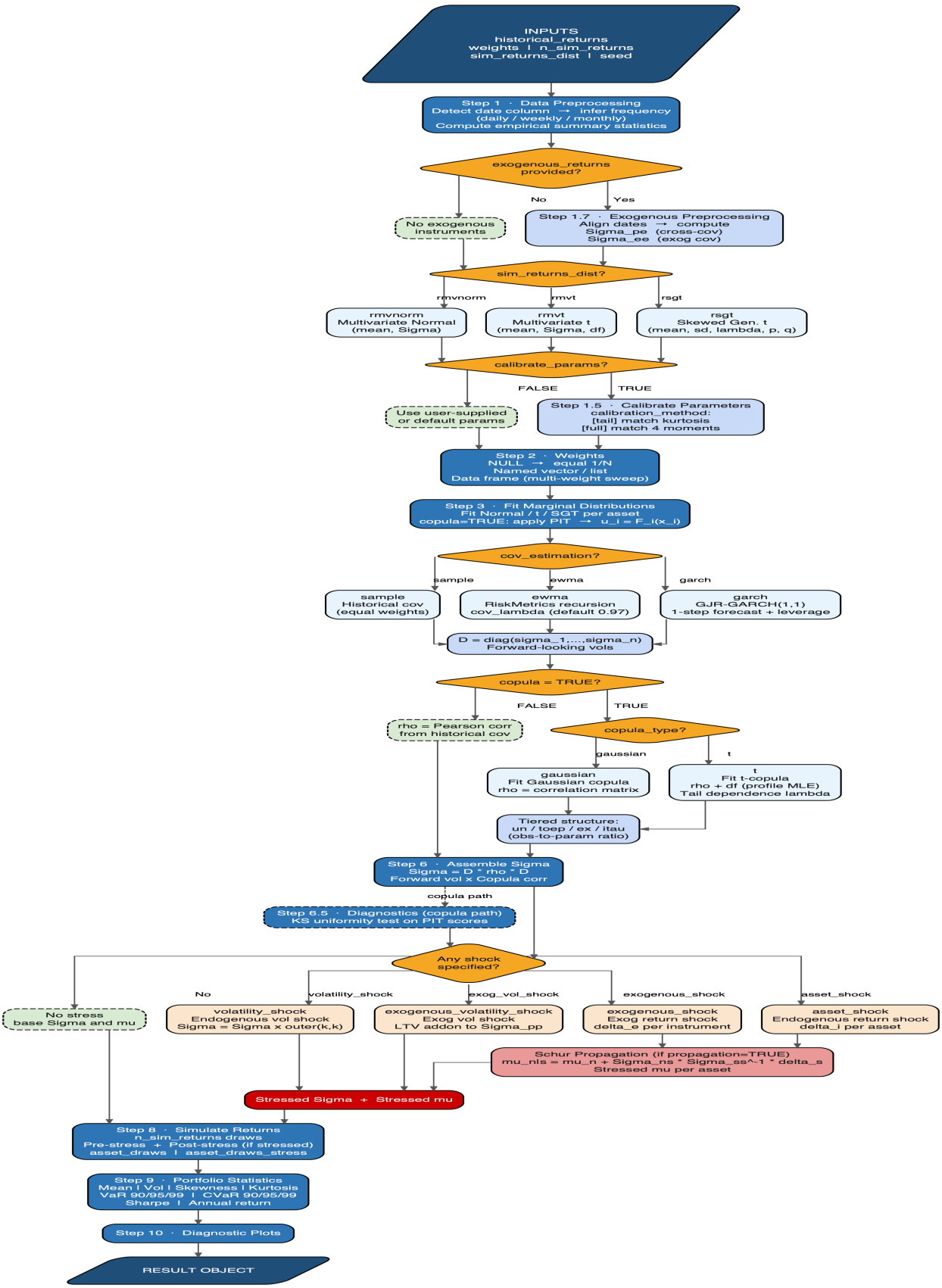

Figure 1: portfolio_risk_simulation() pipeline.

2 Data Preprocessing (Step 1)

2.1 Input Data Frames

The primary input is `historical_returns`, a data frame (or tibble) with one column per asset and one row per observation period. A date column is optional but highly recommended: the function uses it to infer the return frequency automatically.

An optional second data frame, `exogenous_returns`, holds time series for instruments that are *not* in the portfolio - for example a broad equity index, or oil futures price. These instruments are not included in the simulation draws but serve as shock channels in the stress-testing step.

2.2 Automatic Frequency Detection

The function computes the median gap (in calendar days) between consecutive observations and classifies the series as follows:

Table 2: Automatic frequency detection rules

Frequency	Detection Condition	Periods / Year
Daily	Median gap <= 5 days	252
Weekly	Median gap <= 10 days	52
Monthly	All other gaps	12

2.3 Portfolio Summary Statistics

After frequency detection, the function prints per-asset summary statistics and stores them in `result$asset_empirical_summary`.

3 Calibrating Distribution Parameters (Step 1.5)

3.1 Choosing a Return Distribution

The argument `sim_returns_dist` selects the distribution used to simulate future asset returns. Three options are available:

Table 3: Return distribution options

Distribution	Key Parameters	Skew?	Fat Tails?
‘rmvnorm‘	Mean, Covariance	No	No
‘rmvt‘	Mean, Scale, df	No	Yes
‘rsgt‘	Mean, SD, lambda, p, q	Yes	Yes

3.1.1 rmvnorm - Multivariate Normal

The Gaussian distribution is fully characterised by a mean vector and a covariance matrix. It is symmetric and assigns exponentially negligible probability to extreme events.

3.1.2 rmvt - Multivariate Student-t

The t-distribution extends the Normal by adding a degrees-of-freedom parameter ν. The density of a d-dimensional multivariate t with mean μ, scale matrix Σ, and ν degrees of freedom is:

$$f(\mathbf{x}) = \frac{\Gamma\left(\frac{\nu+d}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)(\nu\pi)^{d/2}|\Sigma|^{1/2}}\left(1+\frac{1}{\nu}(\mathbf{x}-\mu)^\top\Sigma^{-1}(\mathbf{x}-\mu)\right)^{-(\nu+d)/2}$$

As ν decreases, the tails become heavier. The t-distribution is **symmetric** - it cannot capture negative skewness.

3.1.3 rsgt - Skewed Generalised t

The SGT distribution (Hansen 1994; Theodossiou 1998) nests several special cases and is parameterised by:

- $\lambda \in (-1, 1)$: **skewness** - negative values indicate left skew (crash-prone)
- $p > 0$: **peakedness** of the central region
- $q > 1/p$: **tail decay** - lower q implies heavier tails

The SGT can simultaneously capture asymmetry and fat tails.

3.2 Calibration Mechanism

When `calibrate_params = TRUE`, the function fits shape parameters to each asset's return history using moment matching. The argument `calibration_method` controls which moments are prioritised during fitting:

- **`"tail"` (default):** Fits parameters by minimising the discrepancy between the model's implied fourth central moment (excess kurtosis) and the empirical excess kurtosis of the return series. Because kurtosis is driven almost entirely by the behaviour of the distribution in its tails, this method effectively forces the shape parameters to reproduce the observed frequency

and severity of extreme returns. The cost is that the fit to the centre of the distribution (mean and variance) is accepted as-is rather than jointly optimised. This may be the preferred setting for VaR and CVaR applications, where tail accuracy outweighs central-moment precision.

- **`"full"`:** Fits parameters by minimising the joint discrepancy across all four moments simultaneously - mean, variance, skewness, and kurtosis - with equal weighting. The optimiser trades off some tail accuracy in exchange for a better overall distributional fit. This may be appropriate when the full shape of the distribution matters, for example when the simulation output is used for scenario generation or distribution forecasting rather than pure tail-risk measurement.

3.3 Reference Asset

The argument `reference_asset` specifies a portfolio asset that serves as the benchmark axis in the pairwise latent-space diagnostic scatter plots. The reference asset appears on the y-axis, with each other asset on the x-axis, so departures from the diagonal can reveal non-linear or asymmetric dependence.

4 Weights Processing (Step 2)

4.1 Weight Input Formats

The `weights` argument accepts three formats:

Table 4: Weight input formats

Format	Description
‘NULL‘ (default)	Equal weights 1/N across all assets. Appropriate for exploratory analysis.
Named numeric vector	Fixed allocation, e.g. ‘c(Equities = 0.6, Bonds = 0.3, Gold = 0.1)‘.
Data frame / matrix	Each row is a distinct allocation. Enables allocation comparison or weight sweeps.

4.2 Number of Simulation Draws

The argument `n_sim_returns` controls the number of joint return draws. When `NULL` (default), the number of simulations conducted is 10000.

5 Fitting Marginal Distributions (Step 3)

5.1 Marginal vs. Joint Distributions

A **marginal distribution** describes the behaviour of a single asset in isolation. A **joint distribution** describes the simultaneous behaviour of all assets and requires specification of the dependence structure.

The function separates these two problems. Step 3 fits a marginal distribution to each asset individually. Step 5 captures the dependence structure via a copula. This is the **copula approach** and is substantially more flexible than fitting a joint distribution directly.

5.2 Without a Copula (`copula = FALSE`)

When `copula = FALSE`, the function simulates from a parametric multivariate distribution in one shot. The dependence structure is implicit in the covariance matrix and the marginals are constrained to be of the same distributional family across all assets.

- **`rmvnorm`:** A standard multivariate Normal draw using the Cholesky decomposition of Σ.
- **`rmvt`:** A multivariate Student-t draw. All assets share the same degrees-of-freedom parameter, so each asset's tail heaviness is identical. The dependence structure is governed by Σ, with stronger co-movement in the tails than the Gaussian.
- **`rsgt`:** Each asset is drawn from its own fitted SGT marginal, and the draws are coupled through a Gaussian copula implied by the Cholesky decomposition of Σ. This is a middle ground: it allows each asset to have a distinct skewness and tail shape (unlike `rmvt`, which imposes a single df across all assets), but the dependence structure remains Gaussian - there is no excess tail co-dependence beyond what Σ implies. To obtain both heterogeneous marginals and non-Gaussian dependence, use `rsgt` with `copula = TRUE`.

5.3 With a Copula (`copula = TRUE`)

A **copula** is a function that couples arbitrary marginal distributions into a joint distribution whilst preserving the chosen marginals exactly. By **Sklar's theorem** (1959), any joint continuous distribution F can be written as:

$$F(\mathbf{x}) = C(F_1(x_1), F_2(x_2), \dots, F_d(x_d))$$

where F_i are the marginal CDFs and $C : [0,1]^d \to [0,1]$ is the copula. This means we are free to choose the best marginal distribution for each asset independently, and specify the dependence structure separately.

When `copula = TRUE`, Step 3 transforms each asset's fitted residuals into uniform $[0,1]$ scores using the **Probability Integral Transform (PIT)**:

$$u_i = F_i(x_i)$$

If the marginal model is correctly specified, these scores are Uniform$(0,1)$. They are then passed to Step 5 where a copula is fitted.

5.4 Positive Definite Matrices

A covariance matrix must be **positive definite (PD)**: all eigenvalues $\lambda_k > 0$. This guarantees that any portfolio's variance $\mathbf{w}^\top \Sigma \mathbf{w} > 0$. When the raw Kendall τ matrix is not PD (common with many assets), the function applies Higham's (2002) `nearPD()` algorithm:

$$\Sigma^* = \arg\min_{M \in \mathcal{S}^+} \|M - \Sigma\|_F$$

where $\mathcal{S}^+$ is the cone of symmetric PD matrices.

6 Forward-Looking Volatilities (Step 4)

Historical sample volatility treats every observation as equally informative about future volatility. Empirical research shows that equity volatility is **clustered** - periods of high volatility tend to follow one another. A forward-looking estimate that weights recent observations more heavily yields better one-period-ahead forecasts.

The argument `cov_estimation` selects the method. The output feeds directly into the covariance matrix constructed in Step 6.

6.1 Methods Comparison

Table 5: Volatility estimation methods

Method	Argument	Description
Historical (Sample)	‘”sample”‘	Simple; unbiased long-run estimate
EWMA	‘”ewma”‘	Reacts quickly to recent vol; single lambda tuning
GJR-GARCH(1,1)	‘”garch”‘	Captures vol clustering and leverage effect

6.1.1 Historical (Sample) Covariance

The sample covariance matrix $\hat{\Sigma} = \frac{1}{n-1}\sum_{t=1}^{n}(\mathbf{r}_t - \bar{\mathbf{r}})(\mathbf{r}_t - \bar{\mathbf{r}})^\top$ treats every observation in the history as equally informative about future volatility. This is unbiased in the statistical sense - no single time period is favoured - but it is may be unresponsive. A spike in volatility two years ago receives exactly the same weight as last week's observation.

6.1.2 EWMA

The Exponentially Weighted Moving Average covariance weights each observation by an exponentially decaying factor λ^{n-t}, where $\lambda \in (0, 1)$ is the decay parameter (`cov_lambda`):

$$\hat{\Sigma}_t = \lambda\,\hat{\Sigma}_{t-1} + (1-\lambda)\,\mathbf{r}_{t-1}\mathbf{r}_{t-1}^\top$$

Observations decay in influence at rate λ per period. The commonly used RiskMetrics value of $\lambda = 0.94$ (daily) or $\lambda = 0.97$ (weekly) gives a half-life of roughly 11 or 23 periods respectively, meaning the most recent period contributes $(1-\lambda)$ of the total weight while older observations contribute progressively less. Unlike GARCH, EWMA has no mean-reversion to an unconditional level - it simply tracks recent realised volatility.

6.1.3 GJR-GARCH(1,1)

The Glosten-Jagannathan-Runkle GARCH model (1993) adds an asymmetric **leverage effect**: negative return shocks increase future volatility more than positive shocks of equal magnitude:

$$\sigma_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \gamma_1\varepsilon_{t-1}^2\mathbf{1}[\varepsilon_{t-1} < 0] + \beta_1\sigma_{t-1}^2$$

When $\gamma_1 > 0$, a negative shock $\varepsilon_{t-1} < 0$ adds $\gamma_1 \varepsilon_{t-1}^2$ to the next period's variance, whereas a positive shock of the same magnitude does not. This captures the well-documented equity market phenomenon where drawdowns generate more volatility than equivalent rallies. The **unconditional (long-run) volatility** is retrieved via `rugarch::uncvariance()`:

$$\bar{\sigma} = \sqrt{\frac{\omega}{1 - \alpha_1 - \frac{\gamma_1}{2} - \beta_1}}$$

The model returns three quantities per asset: the in-sample conditional volatility path, the unconditional volatility, and the 1-step-ahead forecast - all displayed in the GARCH time-series plot.

7 Fitting the Copula (Step 5)

7.1 Copula Types

By Step 5, each asset's return has been expressed as a uniform $[0, 1]$ PIT score. The copula models the dependence between these uniform residuals - it captures *how assets co-move after their individual distributional shapes have been accounted for*.

Two copula families are available via `copula_type`:

- **Gaussian copula:** Dependence described entirely through a correlation matrix R. There is **no excess co-movement in the tails** - assets are asymptotically independent in the extremes.
- **t-copula:** Extends the Gaussian with a degrees-of-freedom parameter ν_C. Lower ν_C implies stronger **tail dependence**.

7.2 Tiered Copula Structure Selection

The feasibility of fitting an **unstructured** (`"un"`) correlation matrix - where every pairwise correlation is estimated freely - depends on the observation-to-parameter ratio. For n assets, the unstructured copula requires $n(n-1)/2$ free parameters:

Table 6: Tiered copula structure selection

Condition	Structure	Description
obs/params >= 20	Unstructured (‘un‘)	Each pairwise correlation estimated freely
obs/params >= 10	Toeplitz (‘toep‘)	Correlations depend only on lag distance between assets
obs/params >= 5	Exchangeable (‘ex‘)	Single correlation parameter for all pairs
Below 5	Kendall tau inversion	Direct tau-inversion; always PD; df recovered via profile MLE

This tiered strategy is grounded in two related bodies of statistical theory. First, the **curse of dimensionality in covariance estimation** (Stein 1956; Ledoit and Wolf 2004): the sample correlation matrix becomes unreliable - specifically, its smallest eigenvalues collapse toward zero - as soon as the observation-to-parameter ratio n/p is not substantially greater than one. Ledoit and Wolf (2004) formalise this precisely, showing that even with $n/p \to \infty$ the sample estimator is suboptimal, and that shrinkage toward a structured target (such as the identity or an equicorrelation matrix) is always preferable in finite samples. Second, **information criteria** (Akaike 1974; Schwarz 1978) penalise model complexity as $2k$ (AIC) or $k \log n$ (BIC), where k is the number of free parameters. Fitting 78 free correlations (13-asset unstructured copula) from 200 observations would score poorly on either criterion because the likelihood gain does not justify the parameter cost. The tiered structure operationalises these principles pragmatically: at each threshold, the structure with the fewest free parameters that the data can reliably support is selected.

7.3 Warm Starting from Kendall's τ

Before numerical optimisation begins, the function constructs a **warm start**: an initial value for every copula correlation parameter, derived analytically rather than through any iterative search. For the t-copula the formula is:

$$\hat{\rho}_{ij}^{\text{warm}} = \sin\left(\frac{\pi}{2}\tau_{ij}\right)$$

For the Gaussian copula (using Spearman's ρ_S):

$$\hat{\rho}_{ij}^{\text{warm}} = 2\sin\left(\frac{\pi}{6}\rho_{S,ij}\right)$$

What these formulas produce: For every asset pair (i, j), Kendall's τ_{ij} is computed from the raw return data. Plugging it into the formula above yields $\hat{\rho}_{ij}^{\text{warm}} \in [-1, 1]$ - a plausible copula correlation for that pair. This is the method-of-moments estimator (itau) and is in fact the *exact* answer when the fitting method is `itau`. For maximum likelihood fitting (mpl), it serves as the starting point.

How the warm start is used: The $n(n-1)/2$ pairwise warm-start values are assembled into a correlation matrix, which is projected to the nearest positive-definite matrix if needed, and passed as the initial parameter vector to the numerical optimiser (`optim` or `fitCopula`). The optimiser then adjusts this starting matrix along the gradient of the copula log-likelihood until convergence. Because the warm start is already close to the MLE (Kendall's τ and the MLE typically agree to within a few percent), the optimiser needs far fewer function evaluations and is much less likely to terminate at a local optimum. Without a warm start - for example, initialising all correlations at zero - the optimiser must traverse a much longer, more rugged path through the parameter space, and convergence failures are substantially more common.

7.4 Fallback: Kendall τ Inversion

If all tiered fitting attempts fail, the function falls back to a two-stage procedure that abandons joint optimisation of (ρ, ν) in favour of estimating each separately.

Stage 1 - The correlation matrix. The Kendall τ-implied correlation $\hat{\rho}_{ij} = \sin(\pi\tau_{ij}/2)$ is computed for every asset pair and assembled into an $n \times n$ matrix. This is the itau (method-of-moments) estimator and is a valid, consistent estimator of the copula correlation - it is not a fallback in the statistical sense, only in the computational sense. If this assembled matrix is not positive-definite (which can happen because pairwise estimates are not guaranteed to be globally consistent), `nearPD()` projects it to the nearest PD matrix. The result is used directly as $\hat{\rho}$ without any further optimisation.

Stage 2 - The degrees-of-freedom parameter. With $\hat{\rho}$ fixed, ν is the only remaining unknown. Searching over ν alone is a **one-dimensional** problem, which is why it can succeed even when the full joint optimisation over (ρ, ν) failed. The search proceeds in two steps:

- **Coarse grid search:** The profile log-likelihood $\ell(\nu \mid \hat{\rho})$ - meaning the copula log-likelihood evaluated at the data, treating $\hat{\rho}$ as fixed and varying only ν - is computed at the candidate values $\nu \in \{3, 4, 5, 6, 8, 10, 15, 20, 30\}$. The value that maximises the likelihood is identified. The grid is denser at small ν because the log-likelihood surface is most curved there: moving from $\nu = 3$ to $\nu = 4$ changes tail dependence substantially, while moving from $\nu = 20$ to $\nu = 30$ changes almost nothing.
- **Brent refinement:** Given the best grid point as a starting bracket, Brent's method (a root-finding algorithm that combines bisection, secant, and inverse quadratic interpolation) refines the optimum on the continuous interval $[2.1, 50]$. The lower bound of 2.1 rather than 2 avoids

the singularity at $\nu = 2$ where the t-distribution has infinite variance. The upper bound of 50 reflects that a t-copula with $\nu > 30$ is empirically indistinguishable from a Gaussian copula.

The output of this two-stage procedure is a valid copula parameterised by $\hat{\rho}$ (from itau inversion) and $\hat{\nu}$ (from profile MLE), which is then used in all downstream simulation and stress-testing steps as if it had been fitted by the standard tiered procedure.

7.5 Tail Dependence

Tail dependence quantifies the probability that two assets move to extreme values **simultaneously**. For the t-copula with parameter ρ_{ij} and ν degrees of freedom, the (symmetric) tail dependence coefficient is:

$$\lambda = 2\, t_{\nu+1}\left(-\sqrt{\frac{(\nu+1)(1-\rho_{ij})}{1+\rho_{ij}}}\right)$$

where $t_{\nu+1}$ is the CDF of the Student-t with $\nu + 1$ degrees of freedom.

8 Building the Final Covariance Matrix (Step 6)

8.1 $\Sigma = \mathbf{D}\,\rho\,\mathbf{D}$ Decomposition

The final covariance matrix is assembled as:

$$\Sigma = \mathbf{D}\,\rho\,\mathbf{D}$$

where $\mathbf{D} = \mathrm{diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_n)$ contains the **forward-looking standard deviations** from Step 4. The source of ρ depends on the chosen path:

- **`copula = TRUE`:** ρ is the fitted copula correlation matrix from Step 5. It is estimated from historical PIT residuals and reflects the long-run dependence structure after the individual distributional shapes have been accounted for.
- **`copula = FALSE`:** ρ is the Pearson correlation matrix derived from the historical sample covariance - specifically $\rho = \mathbf{D}_h^{-1}\hat{\Sigma}_{\text{hist}}\mathbf{D}_h^{-1}$, where $\mathbf{D}_h$ contains the historical (not forward-looking) standard deviations. The forward-looking diagonal $\mathbf{D}$ then replaces the historical volatilities whilst preserving the historical correlation structure.

In both cases, the separation is deliberate: the diagonal $\mathbf{D}$ supplies forward-looking volatility estimates that are deemed to be suitable and responsive to the current regime, while the off-diagonal structure in ρ reflects the longer-run dependence pattern.

8.2 Scale Matrix for the t-Copula Path

When simulating from a t-copula, the copula correlation matrix must be converted to a **scale matrix**. For $\nu > 2$ degrees of freedom, the covariance equals $\nu/(\nu - 2)$ times the scale matrix. The function applies the inverse correction so that the simulated returns have the covariance Σ constructed above:

$$\Sigma_{\text{scale}} = \frac{\nu - 2}{\nu}\,\Sigma$$

8.3 Interpreting Diagnostics

Condition number. The condition number $\kappa(\Sigma) = \lambda_{\max}/\lambda_{\min}$ measures how sensitive the matrix inverse is to small perturbations in the data. Intuitively, a high condition number means that two or more assets move together so reliably that the matrix is effectively trying to distinguish directions that look nearly identical to it - small numerical errors get amplified into large errors in the inverse, which then propagates into portfolio variance calculations and copula fitting. The thresholds $\kappa < 100$ (well-conditioned) and $\kappa > 1{,}000$ (near-multicollinearity) are conventional rules of thumb drawn from numerical linear algebra (Golub and Van Loan 2013) rather than formal statistical tests. They indicate the order of magnitude of numerical reliability rather than a hard boundary.

Eigenvalues. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ of a covariance matrix have two roles. First, all must be strictly positive for the matrix to be positive-definite (a requirement for valid portfolio variance and for the Cholesky decomposition used in simulation). Second, their *magnitudes* matter: large eigenvalues correspond to directions of high variance in the data (principal components with large explained variance), while small eigenvalues correspond to near-flat directions. A minimum

eigenvalue that is positive but very small (10^{-6})after `nearPD()` correction - indicates that the matrix is barely positive-definite and that the corresponding portfolio direction has negligible variance in the data. This is not an error per se, but it signals that the correlation matrix may be capturing noise rather than genuine structure in that direction, and that results involving that eigendirection (such as the fitted copula for those asset pairs) should be interpreted with caution.

9 Latent Variable Diagnostics (Step 6.5)

9.1 Latent vs. Observed Space

The **latent space** refers to the uniform $[0, 1]$ PIT residuals. If the marginal models are correctly specified, these should be uniformly distributed, and their joint distribution should match the fitted copula.

The pairwise scatter plots display each asset pair in *both* spaces - the observed return space and the latent uniform space - on the same panel. Departures from the expected pattern are informative:

- **Banana-shaped curves** in the observed space: Non-linear relationship that the copula may not fully capture.
- **Corner clustering** in the uniform space: Tail dependence - assets that tend to move to extremes simultaneously.
- **Asymmetric corner density**: Lower-left clustering (joint crashes) without upper-right clustering (joint rallies) signals asymmetric tail dependence.

9.2 KS Uniformity Test

The Kolmogorov-Smirnov (KS) test for each asset tests $H_0 : u_i \sim \text{Uniform}(0, 1)$:

$$D_n = \sup_{u \in [0,1]} |F_n(u) - u|$$

A p-value below 0.05 indicates that the fitted marginal distribution is misspecified for that asset. Results are stored in `result$ks_test`. Common causes of rejection: heavy tails that the model cannot accommodate, bimodal distributions, or severe outliers.

10 Stress Testing (Step 7)

10.1 Overview of Shock Types

The stress-testing framework distinguishes two **sources** (endogenous = portfolio assets; exogenous = non-portfolio instruments) and two **types** (return shock; volatility shock), controlled via four arguments:

Table 7: Stress testing arguments

Argument	Target	Shock Type	Unspecified Default
‘asset_shock‘	Portfolio assets	Return (decimal)	0 per asset
‘volatility_shock‘	Portfolio assets	Vol multiplier k	1 per asset (no change)
‘exogenous_shock‘	Exogenous instruments	Return (decimal)	0 per instrument
‘exogenous_volatility_shock‘	Exogenous instruments	Vol multiplier k	1 per instrument

10.2 Schur Complement Mean Propagation Formula

When `propagation = TRUE`, a return shock to any asset or exogenous instrument is propagated to all other portfolio assets via the **Schur complement conditional mean formula**:

$$\mu_{n|s} = \mu_n + \Sigma_{ns}\,\Sigma_{ss}^{-1}\,\delta_s$$

where:

- subscript s = the **shocked block** (endogenous return-shocked assets $\cup$ exogenous return-shocked instruments)
- subscript n = non-shocked portfolio assets
- δ_s = vector of return shocks applied to block s
- Σ_{ns} = cross-covariance between non-shocked and shocked assets

This is the conditional expectation of the non-shocked assets given the shocked assets - a standard result from multivariate normal theory. The formula is valid because Σ_{ss} is positive definite, guaranteeing invertibility.

10.3 Volatility Shock Mechanics

10.3.1 Endogenous Volatility Shock

An endogenous volatility shock applies a scalar multiplier k_i to asset i's row and column of the covariance matrix. Applied simultaneously across all shocked assets:

$$\Sigma_{\text{stressed}} = \Sigma \odot (\mathbf{k}\,\mathbf{k}^\top)$$

where $\mathbf{k}$ is the vector of multipliers ($k_i = 1$ for unshocked assets) and $\odot$ denotes the Hadamard (element-wise) product. This means that volatilities are scaled multiplicatively: asset i's own variance is multiplied by k_i^2, and the covariance between assets i and j is multiplied by $k_i \times k_j$. The pairwise correlations between assets are therefore preserved exactly - applying a volatility shock to

asset i does not change how correlated it is with asset j; it only changes how large the absolute co-movement is. Intuitively, doubling asset i's volatility ($k_i = 2$) makes its returns twice as large in magnitude but leaves its directional relationship with every other asset unchanged. The portfolio-level effect is that any position exposed to asset i sees its variance increase by a factor of $k_i^2 = 4$, and its covariance with positions exposed to asset j increases by $k_i k_j = 2$ (assuming $k_j = 1$).

10.3.2 Exogenous Volatility Shock

An exogenous volatility shock to instrument e transmits additional variance to the portfolio via the **Law of Total Variance (LTV)**:

$$\Sigma_{pp}^{\text{stressed}} = \Sigma_{pp} + \sum_e (k_e^2 - 1)\, \frac{\sigma_{pe}\, \sigma_{pe}^\top}{\sigma_{ee}}$$

where $\sigma_{pe} = \Sigma_{pe}[\cdot, e]$ is the column of cross-covariances with exogenous asset e, and $\sigma_{ee} = \Sigma_{ee}[e, e]$ is its variance. Key properties:

- $k_e = 1 \Rightarrow$ addon $= 0$: Covariance unchanged
- $k_e > 1 \Rightarrow$ addon > 0: Covariance widens
- Return shock δ_e: has **no effect** on covariance

A -10% return shock to an exogenous instrument (e.g. SPY) causes a large leftward shift in the portfolio mean distribution through the Schur propagation, but has **zero effect** on portfolio volatility. Volatility only changes when the exogenous vol multiplier $k_e \neq 1$.

10.4 Propagation vs. No Propagation

When `propagation = FALSE`, the Schur complement step is skipped and shocked assets are modified in isolation - no conditional mean update flows to other portfolio assets. This is a **standalone** stress: how much does this asset alone contribute to the stressed portfolio return, ignoring correlations with other assets?

Comparing the stressed portfolio distribution with propagation *on* versus *off* quantifies the **propagation effect** - the incremental risk attributable to inter-asset correlations, above and beyond the direct effect of the shock.

10.5 Stress Scenario Matrix

The table below covers all outcomes arising from the four binary shock dimensions (endo return, endo vol, exog return, exog vol). Scenarios with identical mean and covariance mechanics are grouped.

Table 8: Complete stress scenario matrix: 16 distinct behavioural patterns. Counts of shocked assets affect magnitude only, not type.

Scenario	Endo R	Endo V	Exog R	Exog V	Effect on Mean	Effect on Covariance
0: Baseline	0	0	0	0	Unchanged	Unchanged
A1	>=1	0	0	0	Shocked: mu_i + delta_i; others: Schur propagation	Unchanged

A2	0	>=1	0	0	Unchanged	Sigma odot outer(k,k)
A3	>=1	>=1	0	0	Schur from endo return-shocked block	Sigma odot outer(k,k)
B1	0	0	>=1	0	All portfolio: mu_p + Sigma_pe Sigma_ee^{-1} delta_e	Unchanged
B2	0	0	0	>=1	Unchanged	LTV addon: Sigma_e [(k_e^2-1) sigma_pe sigma_pe' / sigma_ee]
B3	0	0	>=1	>=1	All portfolio: Schur from exog return block	LTV addon (independent of return shock)
C1	>=1	0	>=1	0	Shocked endo + all: compound Schur (endo + exog block)	Unchanged
C2	>=1	0	0	>=1	Endo Schur propagation only	LTV addon [exog vol]
C3	0	>=1	>=1	0	All portfolio: Schur from exog return block	Sigma odot outer(k,k) [endo vol]
C4	0	>=1	0	>=1	Unchanged	Sigma odot outer(k,k) + LTV addon - additive
C5	>=1	>=1	>=1	0	Compound Schur: endo + exog return blocks	Sigma odot outer(k,k) [endo vol]
C6	>=1	>=1	0	>=1	Endo Schur propagation only	Sigma odot outer(k,k) + LTV addon - additive
C7	>=1	0	>=1	>=1	Compound Schur: endo + exog return blocks	LTV addon [exog vol only]
C8	0	>=1	>=1	>=1	Exog Schur propagation only	Sigma odot outer(k,k) + LTV addon - additive
C9	>=1	>=1	>=1	>=1	Full compound Schur: endo + exog return blocks	Sigma odot outer(k,k) + LTV addon - additive

11 Simulating Returns (Step 8)

11.1 Simulation Inputs

The simulation step draws `n_sim_returns` joint return vectors from the fitted distribution. Key inputs from prior steps:

- **Marginal parameters** (Step 3): distribution family and shape parameters for each asset
- **Covariance matrix** Σ (Step 6): forward-looking variance and copula-implied correlation
- **Stressed mean and covariance** (Step 7, if applicable): replaces the base inputs when a stress scenario is active
- **Weight matrix** (Step 2): aggregates simulated asset returns into portfolio-level returns

For the copula path (`copula = TRUE`), simulation proceeds in two stages:

1. Draw $\mathbf{u} \sim C(\rho)$ from the fitted copula (uniform scores)
2. Apply the inverse CDF: $x_i = F_i^{-1}(u_i)$ using each asset's fitted marginal

11.2 Sanity Check: `sim_vol / target_vol`

After simulation, the function computes the ratio of the realised simulation volatility to the target volatility $\sqrt{\Sigma_{ii}}$. A ratio close to 1.0 confirms the simulation is consistent with the fitted covariance.

For strongly skewed distributions (e.g. SGT with large $|\lambda|$), this ratio can deviate from 1.0 because skewness affects the sample standard deviation even when the theoretical variance is correct. A ratio in the range $[0.95, 1.05]$ is considered acceptable.

11.3 Simulation Results Summary

The simulation output includes the following statistics, accessible via `result$portfolio_stats`:

Table 9: Portfolio risk statistics computed from simulation

Statistic	Description
Mean Return	Expected portfolio return over the period
Volatility	Standard deviation of portfolio returns
VaR 90/95/99%	5th/1st percentile of the simulated return distribution
CVaR 95% (Expected Shortfall)	Expected loss conditional on return below VaR threshold
Skewness	Third standardised central moment of portfolio returns
Excess Kurtosis	Fourth standardised central moment minus 3
Sharpe Ratio (Annual)	Annualised (mean / vol) x sqrt(periods per year)

12 Statistics and Diagnostic Plots (Steps 9–10)

12.1 Accessing Results

The function returns a named list with three top-level components:

- `result$portfolio_stats`: Data frame of portfolio risk statistics (pre- and post-stress)
- `result$plots$portfolio_diagnostics`: List of ggplot/patchwork objects for model diagnostics
- `result$stressed_plots`: Stress-scenario visualisations (populated only when shocks are specified)

Additional result fields are summarised in the table below:

Table 10: Key result object fields

Field	Contents
result$asset_empirical_summary	Per-asset empirical mean, SD, skewness, kurtosis, min, max
result$asset_draws	n_sim x n_assets data frame of pre-stress per-asset draws
result$asset_draws_stress	n_sim x n_assets data frame of post-stress draws; NULL if unstressed
result$exogenous_asset_empirical_summary	Empirical summary for exogenous instruments; NULL if none
result$exogenous_historical_correlation	Portfolio x exogenous Pearson correlation data frame; NULL if no exog
result$ks_test	KS test: Asset, KS_stat, p_value, Flag; NULL if copula = FALSE
result$stress_impact_summary	7-row data frame: Metric / Pre_Stress / Post_Stress / Change; NULL if unstressed
result$portfolio_stats	14-row data frame: all portfolio risk statistics pre and post stress
result$calibrated_params	Calibrated distribution shape parameters per asset; NULL if not calibrated
result$final_covariance	Final n x n forward-looking covariance matrix (Sigma)
result$copula_correlation	Fitted copula correlation matrix (rho); NULL if copula = FALSE
result$tail_dependence	Tail dependence matrix (lambda); NULL if Gaussian copula or copula = FALSE

12.2 Diagnostic Plot Inventory

Table 11: Portfolio diagnostic plot inventory

Plot Name	Description and Insights
‘asset_distributions‘	Histogram of each asset’s return with fitted marginal overlay. Spot bimodal distributions, outliers, or poor marginal fits.
‘portfolio_distribution‘	Histogram of the simulated portfolio return. Shows the full shape, not just point estimates.
‘qq_plot‘	QQ plot vs. Normal reference. Points above the diagonal in the left tail indicate heavier left-tail risk.
‘historical_correlation‘	Pearson correlation heatmap. Identifies clusters of highly correlated assets and natural diversifiers.

‘copula_correlation‘	Copula correlation matrix alongside its difference from historical Pearson. Divergences indicate non-linear dependence.
‘covariance_matrix‘	Heatmap of the final forward-looking covariance (Sigma), labelled with sigma on diagonal and rho off-diagonal.
‘time_series‘ (GARCH only)	Multi-panel canvas (4 x ceil(n/4)) of conditional vol, raw returns, unconditional vol, and 1-step forecast per asset.
‘pairwise_diagnostics‘	Pairwise scatter in observed and latent spaces for each asset vs. the reference asset. Reveals non-linear dependence.
‘tail_dependency‘	Tail dependence bar chart per asset pair + copula rho vs. Kendall tau-inversion rho scatter, coloured by |Delta rho|.

12.3 Stress-Scenario Plots

When stress scenarios are active, the following additional plots are produced:

- **`portfolio_distribution_comparison`:** Overlaid pre-stress and post-stress portfolio return distributions, with VaR and CVaR markers. The central visualisation for communicating stress impact to stakeholders.
- **`exogenous_propagation` (Scenario B):** Transmission of the exog shock from instruments to each portfolio asset, scaled by the implied beta β_{pe}.
- **`propagation_effects` (Scenario A):** Conditional mean shift per asset, broken down into direct (shocked asset) and propagated (correlated assets) components.
- **`mixed_shock_effects` (Scenario C):** Compound effect decomposition: direct endo + propagated endo + propagated exog.

13 Quick-Reference Call Examples

13.1 Minimal Call

```
result <- portfolio_risk_simulation(
  historical_returns = my_returns
)
```

13.2 Full SGT + GARCH + t-Copula

```
result <- portfolio_risk_simulation(
  historical_returns  = my_returns,
  sim_returns_dist    = "rsgt",
  cov_estimation      = "garch",
  calibrate_params    = TRUE,
  calibration_method  = "tail",
  copula              = TRUE,
  copula_type         = "t",
  n_sim_returns       = 10000,
  seed                = 42
)
```

13.3 Endogenous Stress Scenario

```
result <- portfolio_risk_simulation(
  historical_returns = my_returns,
  sim_returns_dist   = "rsgt",
  cov_estimation     = "garch",
  copula             = TRUE,
  asset_shock        = list(Energy = -0.15, Finance = -0.10),
  volatility_shock   = list(Energy = 1.5),
  propagation        = TRUE
)
```

13.4 Exogenous Shock via SPY

```
result <- portfolio_risk_simulation(
  historical_returns          = my_returns,
  exogenous_returns           = exog_df,
  exogenous_shock             = list(SPY = -0.10),
  exogenous_volatility_shock  = list(SPY = 2.0),
  propagation                 = TRUE
)
```

14 References

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